Large Language Models in Embodied and Mobile Robotics: A Systematic Review of Neuro-Symbolic Integration and Autonomous Error Recovery

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***Abstract*—Large Language Models (LLMs) can translate high-level commands into robot plans but may generate physically invalid actions and often lack execution-aware recovery. We systematically reviewed 46 empirical studies (Scopus, IEEE Xplore, ACM Digital Library; January 2020 to 3 November 2025; plus a supplementary open-access retrieval round). We classify them as direct generation (39.1%), static neuro-symbolic translation (39.1%), formal verification without autonomous feedback (8.7%), or runtime plan adaptation and recovery (13.0%). The six studies in the last category operate in restricted domains: three combine runtime recovery with formal verification, one combines lightweight symbolic state with human-executed recovery, and two recover without formal symbolic verification. Among the 46 included studies, none demonstrates semantic diagnosis of physical execution failures, runtime updates to formal constraints, and autonomous replanning in dynamic, unstructured environments. We characterize this bounded gap and derive four candidate requirements for closed-loop neuro-symbolic recovery.**

***Index Terms*—Action hallucination, embodied robotics, large language models, mobile robotics, neuro-symbolic architecture, planning domain definition language (PDDL), systematic literature review.**

I. Introduction

Embodied robots are increasingly expected to carry out instructions in homes, hospitals, and other settings that are less predictable than industrial workcells [2]. A robot asked to "take this medicine to room three" may find that a door is locked or a corridor is blocked after planning has begun.

Large Language Models (LLMs) can translate natural-language requests into robot plans [3]. Yet a plan may be syntactically valid while violating the robot's physical constraints, a problem often described as action hallucination [4]. Grounding and checking a plan before execution help, but they cannot by themselves resolve a failure that occurs after the robot starts acting. The execution system also needs a way to report the changed state and request a revised plan.

We reviewed 46 empirical studies to examine how LLM-enabled robotic systems represent the physical world, check proposed actions, and respond to execution failures. Earlier reviews discuss language grounding [12], LLM–robot integration [13], LLM planning [14], and formal planning models [15]. Our focus is narrower: we examine how a system moves from an initial language-generated plan to monitoring, diagnosis, and revision during execution.

The paper contributes (i) a four-group taxonomy based on symbolic structure and runtime feedback; (ii) a reproducible mapping of 46 studies, including 11 added after supplementary full-text retrieval; (iii) an assessment of the evidence for semantic failure diagnosis, formal-model updating, and autonomous replanning; and (iv) four candidate requirements for closing the recovery loop.

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II. Methodology

We followed the PRISMA 2020 reporting guidelines [1] to document the search and selection of empirical studies. The review covers LLMs and vision-language models (VLMs) in embodied robotics, with particular attention to grounding, planning validity, and execution-error handling across mobile navigation, manipulation, and embodied simulation.

A. Search Strategy

The search covered Scopus (99 records), IEEE Xplore (45), and ACM Digital Library (39), with publications dated January 2020 to 3 November 2025. Nine database-specific queries combined language-model, robotics, grounding, planning, verification, and recovery terms. The final search was performed on 3 November 2025, so coverage of November is incomplete. The databases returned 183 records before deduplication. The supplement and companion repository archive the queries, decisions, and extraction data [58] (release v1.1). Publisher PDFs are not redistributed. Retrieval of open-access versions in August 2026 supplemented the original search without extending its publication window.

B. Inclusion and Exclusion Criteria

We use embodied robotics as the umbrella scope, including mobile, mobile-manipulation, fixed-base, tabletop, humanoid, and embodied-simulation platforms; mobile navigation was not required.

Inclusion Criteria (IC):

- **IC1:** Proposes or utilizes LLMs or vision-language models (VLMs) for robot planning, control, task decomposition, or semantic interpretation of robot task state (e.g., execution logs).
- **IC2:** Presents empirical validation in simulation, physical hardware, or datasets/offline benchmarks.

- **IC3:** Addresses robotic grounding or safety/reliability mechanisms, hallucination mitigation, formal planning or verification, or execution-failure handling.

Exclusion Criteria (EC):

- **EC1:** Does not involve an embodied robotic system or an embodied simulation of a robot task.
- **EC2:** Purely theoretical work without implementation or experimental validation.
- **EC3:** Does not address any of the grounding, safety, verification, mitigation, or recovery mechanisms in IC3.

C. Study Selection Process

We removed 35 duplicates from the 183 records and screened 148 units by title and abstract. After 82 exclusions, 66 full-text reports were sought. Institutional access initially failed for 28 of these reports, leaving 38 assessed and 35 included in the original review. In August 2026, we identified possible open-access versions for 18 of the previously unavailable reports and obtained 17 full texts. Applying the documented eligibility criteria yielded 11 further inclusions and six exclusions; one report remained inaccessible. The final flow is therefore 55 assessed, nine excluded after full-text assessment, 11 not retrieved, and **46 studies included** (Fig. 1).

The early screening archive contains 147 bibliographic records plus the original `AdjunctProceedings*` aggregate row, which accounts for 17 grouped proceedings records. Details and exclusion reasons are in the supplement. Screening, extraction, and classification were performed by the first author. The supplementary decisions followed written rules fixed for that retrieval round; no independent second-reviewer series is available, and we report no inter-rater agreement statistic.

III. Taxonomy of LLM-Robot Integration

We classified the studies along two dimensions: the degree of symbolic structure used to represent or check a plan, and whether execution feedback can trigger an autonomous change in the planning process. The first dimension distinguishes methods without an external symbolic model, lightweight representations such as ontologies and scene graphs, and explicit formal planning or verification models. The second separates preparation or checking before execution from adaptation in response to runtime observations. This distinction follows the established separation between plan validation and execution monitoring [45].

These axes were selected for the review because action hallucination and execution failure pose different problems. They are an analytical choice, and a taxonomy based on the LLM's role or the robot platform could group studies differently. In the original 35-study corpus, all three recovery systems also used formal models. Supplementary retrieval found recovery in systems with lighter symbolic structure or none, so runtime recovery and formal-model use are independent dimensions. Fig. 2 shows the resulting six cells and the four mutually exclusive groups: A=18, B=18, C=4, and D=6.

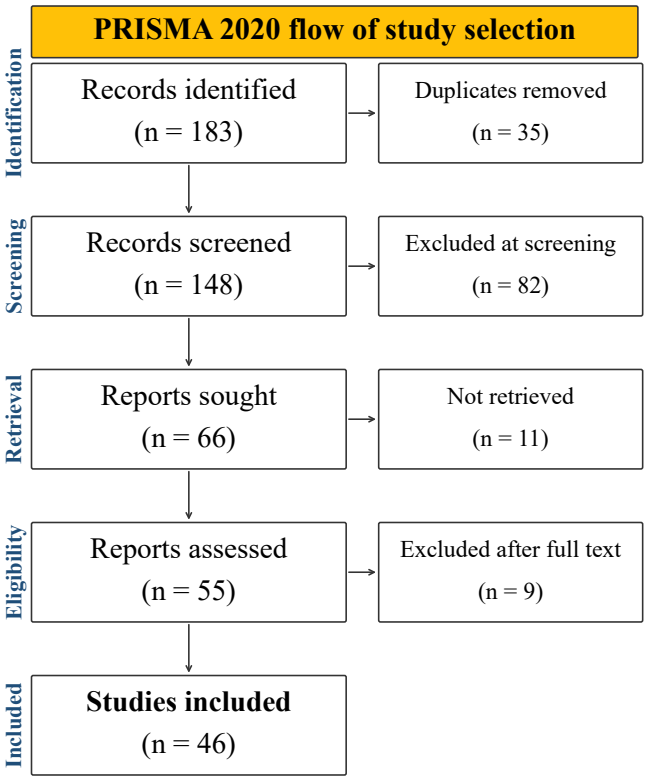


Fig. 1. PRISMA 2020 flow diagram of the study selection process.

A. Group A: Direct Generation (End-to-End)

Group A contains 18 studies (39.1%). In these systems, a language model generates robot actions, trajectories, or task decisions without an external symbolic verifier. For example, companion-robot interaction [17] and driving [42] use learned or prompted representations to select behavior; the generated actions are not checked against an explicit formal model. Further examples and extraction records are in the supplement.

B. Group B: Static Neuro-Symbolic Translation

Group B also contains 18 studies (39.1%). Here, an ontology, scene graph, digital twin, action schema, or other structured representation constrains the model's output [25], [37], [49]. The extracted evidence does not establish autonomous recovery from physical execution failures.

Boundary cases clarify this classification. OceanPlan [22] revises hierarchical task plans at the mission level, without formal verification or failure semantics; Merlo et al. [7] revise actions with a person in the loop; NVP-HRI [49] checks and revises generated trajectories before physical execution. None receives feedback from runtime failure; classification decisions are in the supplement.

C. Group C: Formal Verification Without Autonomous Feedback

Four studies (8.7%) use an explicit formal model without demonstrating autonomous recovery after an execution failure.

Chalvatzaki et al. [31] translate language into PDDL subproblems solved by Fast Downward. Patton et al. [27] include unrealizability checking. Kanazawa et al. [53] plan cooking actions in PDDL, and González-Santamarta et al. [56] combine PDDL-based planning with offline interpretation of robot execution logs. None closes an autonomous execution-recovery loop; use of PDDL evidences a formal representation but does not alone guarantee physical safety or plan correctness.

D. Group D: Runtime Plan Adaptation and Recovery

Group D contains six studies (13.0%). Their common feature is an execution-aware mechanism: a change in observed state, an unmet precondition, or planning uncertainty can trigger a revised plan or an escalation action. The mechanisms are not interchangeable. Zhang et al. [5] check expected state against PDDL preconditions, while Capitanelli and Mastrogiovanni [6] monitor state and preconditions during manipulation. Wang et al. [11] use LTL-based planning under uncertainty and can issue alternative high-level subtasks, although they assume error-free execution of the low-level skills.

The three further studies demonstrate why recovery must be coded independently of formality. PaLM-E [48] uses visual feedback to detect skill failures and trigger replanning. Tziafas and Kasaei [52] repeat grounding and grasp planning after a grasp fails or becomes blocked. Neither uses a formal symbolic verifier. RONAR [54] maps failure states into a runtime scene graph and supports escalation to user queries or teleoperation. The physical recovery remains human-executed; we classify the automatic escalation mechanism as runtime adaptation, not as autonomous robot replanning.

TABLE I
GROUP D STUDIES: RECOVERY MECHANISM VERSUS EMPIRICAL DEMONSTRATION.

Study	Recovery trigger	Validation	Recovery metrics
Zhang et al. (2023)	PDDL precondition check with self-reflection	Hybrid	None (task success only)
Capitanelli et al. (2024)	Post-action state and precondition monitoring	Simulation (Baxter)	None (plan validity, time)
Wang et al. (2025)	Planning uncertainty; infeasible-subtask escalation	Hybrid	Partial (help rate, mission success)
PaLM-E (2023)	Prompted skill-failure detection in the control loop	Hybrid	Failure-detection F1 (0.91)
Tziafas & Kasaei (2024)	Failed grasp triggers re-planning	Hybrid	Recovery ablation (22/8%)
Wang et al. (2024)	Failure state → query-user/teleoperation (human-executed)	Real	Failure capture rate; explanation accuracy

	No autonomous feedback	Autonomous feedback
No formal verification	**A = 18** direct generation	**D = 2** runtime recovery
Lightweight symbolic	**B = 18** static integration	**D = 1** runtime recovery
Formal model	**C = 4** formal checking	**D = 3** runtime adaptation

Fig. 2. Taxonomy matrix of the 46 studies by symbolic structure and runtime feedback. Group D spans all three levels: PaLM-E and open-world grasping recover without formal symbolic verification (row 1); RONAR (row 2) escalates autonomously but executes physical recovery through a human; the three earliest Group D studies use formal models (row 3). Runtime recovery and verification formality are independent dimensions.

Table I separates the reported trigger from the form of experimental evidence. The first two studies describe plan revision or state checking but report only task- or plan-level measures. The other four provide metrics related to detecting failures, requesting help, or testing a recovery component. An F1 score for failure detection, for example, is not a recovery-success rate; an escalation measure also differs from an autonomous repair. None of the six reports time to recovery, and none evaluates all the dynamic-environment conditions used here (Supplement, Table C1–C3).

E. Counter-Example Audit

Five prominent systems outside the database corpus were audited as counterexamples; each lacks at least one of formal verification or autonomous runtime recovery (Inner Monologue, SayCan, Code-as-Policies, LLM+P, Reflexion; full table in the supplement). Detailed decisions are available in the companion repository [58].

We examine interaction modalities (RQ1), LLMs and mitigation strategies (RQ2), verification mechanisms (RQ3), validation settings (RQ4), and reported technical challenges (RQ5).

IV. Analysis of Research Questions

We extracted data from the 46 studies. Interface and validation-setting categories are mutually exclusive, whereas model families and reported challenges may have multiple labels per study, so totals use different counting rules. All cross-study summaries are descriptive; we make no inferential or causal claims from these frequencies.

A. Interfaces and Perception (RQ1)

Text remains the most common interface in the mapped systems. Unimodal inputs account for 71.7% of the corpus: text in 45.7%, voice in 17.4%, and gestures, gaze, or buttons in 8.7%. Another 23.9% accept multimodal input, while 4.3% do not specify an interface (full counts in the supplement). Allgeuer et al. [9], for example, combine speech recognition with open-vocabulary object detection.

B. LLM Ecosystems and Hallucination Mitigation (RQ2)

The GPT family appears in 30 of 46 studies (65.2%), followed by LLaMA in 11 and Mistral in two. These are study-level, multi-label counts, so family totals need not sum to 46 (ranking in the supplement).

The studies address hallucination in different ways. Some adapt models using domain-specific data [6]; others retrieve sensor or scene information to ground generated actions [4]. Structured reasoning prompts can enumerate constraints [10], [21], while constrained decoding limits output forms [41]. Agentic reflection and additional checking agents are also reported [36], [39]. These methods do not alone demonstrate recovery from physical failures.

C. Verification Strategies (RQ3)

Seven studies (15.2%) use an explicit formal planning or checking model. Examples include PDDL-based planning [5], [6], [31], [53], [56], LTL with conformal prediction [11], and unrealizability checking [27]. Other systems use ontologies, digital twins, scene graphs, or memory to provide less formal structure [25], [35], [37].

We distinguish Level 1 (no formal verification, 20 studies), Level 2 (lightweight symbolic consistency, 19), and Level 3 (explicit formal planning or checking model, seven). These levels describe the method's representation or checking mechanism, not a ranking of empirical safety: formal checks provide guarantees only with respect to what the model and specification encode, and operators may omit obstacles that appear during execution. This is why formality is coded independently from execution feedback.

D. Experimental Validation (RQ4)

Evaluation settings differ substantially across the corpus. Nine studies (19.6%) validate exclusively in simulation, 19 (41.3%) on physical hardware, ten (21.7%) use hybrid evaluations, and eight (17.4%) rely on datasets or offline benchmarks ("Other" in the supplement). These labels describe where a system was tested, not whether it handled execution errors: hardware experiments may report task completion without isolated failure cases, and offline benchmarks need not involve robot execution. Task Success Rate is the most common outcome, while direct safety and recovery measures are less consistently reported.

E. Open Technical Challenges (RQ5)

We coded challenges reported by the authors in their limitations and future-work sections (Fig. 3). Action hallucination was mentioned in 25 studies (54.3%), latency in 17, and generalization in 16. Complexity and physical constraints appeared in nine studies each, misinterpretation in seven, and bias, context-window limitations, execution/grounding errors, and instruction ambiguity in four each. A study can mention several challenges. These figures are counts of reported concerns, not observed failure frequencies or measures of how often robots fail in practice. The single-coder process also limits confidence in borderline labels.

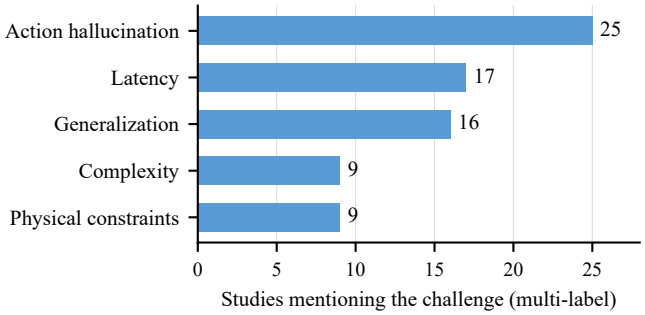


Fig. 3. Five most frequent technical challenges reported by the reviewed studies (RQ5). Categories are multi-label and count mentions in authors' limitations and future-work sections, not observed failure frequencies; the full Pareto chart is in the supplement.

Only six of the 46 studies (13.0%) have the runtime recovery or escalation mechanisms assigned to Group D. Two lack formal symbolic verification, one relies on a human to execute recovery, and none is evaluated against the full combination of environmental and recovery requirements defined below.

V. Discussion and Research Gap

The mapping points to three recurring problems. First, an LLM may produce a plan that is checked once and then executed without a route for reporting changes in the world. Second, a failure may remain a low-level status or motor error without a description suitable for revising the task plan. Third, a formal planner needs an updated model: rerunning it with stale predicates may reproduce the same invalid behavior. LLM inference latency complicates the interaction between execution and deliberation [16]. Fast reactive safety functions should therefore remain independent of LLM calls.

To examine deployment conditions, we defined a *dynamic, unstructured environment* by three observable properties: (C1) obstacles can appear unexpectedly during execution, (C2) task-relevant state can change without warning and invalidate the initial model, and (C3) no complete prior map or object inventory is available. None of the six Group D studies demonstrates all three (Supplement, Table C1–C3). The formal-model systems use more structured environments or assume reliable low-level skills. PaLM-E and open-world grasping handle feedback in bounded laboratory tasks without updating a formal model. RONAR interprets failure states and provides escalation, with recovery executed by a person. Classical planners already support monitoring and plan repair [45], [46]; the uncovered combination in this review is LLM-mediated failure diagnosis, formal-state updating, and autonomous replanning under these deployment conditions.

Four *candidate design requirements* follow from the gaps documented in the corpus. R1 is an error interpreter that converts execution failures into structured semantic descriptions, so a blocked corridor or failed grasp can affect the next planning decision. R2 updates the relevant formal state or constraints before replanning; an unchanged model cannot represent a newly discovered obstacle. R3 confines the LLM

to semantic interpretation while a deterministic planning or checking component handles formal constraints and a separate reactive controller handles immediate safety. R4 specifies recovery and escalation options, such as retry, reroute, replan, request help, and abort, with explicit conditions for choosing among them. The traceability of these proposals to corpus evidence is recorded in the supplement. They are design hypotheses, not demonstrated outcomes, and overlap with architecture work pursued by the authors.

Human escalation remains appropriate when failures are severe, ambiguous, or outside the system's validated capabilities. A recovery experiment should record the observed trigger, failure diagnosis, state update, revised plan, and final task outcome. A retry followed by task completion differs from an operator correcting the robot's state, even if both experiments report success. Recovery success, recovery time, intervention frequency, and safety violations would help distinguish these outcomes. Future comparisons require independent coding and explicit recovery metrics.

VI. Threats to Validity

Selection and retrieval: We searched three bibliographic databases and did not include unpublished work systematically. Of 66 reports sought, 28 were initially inaccessible; the supplementary retrieval identified 18 possible open-access versions and yielded 17 texts for assessment. Eleven reports remain unexamined. The negative finding concerns the 46 included studies, and features absent from inaccessible reports cannot be ruled out. The original grouped screening decision is documented separately from individual full-text exclusions; this historical aggregate limits reconstruction of the earliest screening stage at the level of individual title/abstract decisions, though grouped records and total counts are archived.

Coverage: The final database search ended on 3 November 2025. The August 2026 retrieval round accessed versions of records already identified; it did not constitute an updated search for later publications. Work indexed or published after the cutoff may change the mapped distribution.

Screening and extraction: The first author screened, extracted, and coded the studies, including adjudication of borderline decisions. Criteria for the supplementary round were documented before that round was coded, with evidence retained for each case. The initial review was not independently double-screened and no auditable second-reviewer coding series exists, so we do not report Cohen's κ or agreement percentages.

Classification and researcher bias: Verification formality and autonomous feedback were selected as analytical dimensions. Their choice reflects the intended research question, and different dimensions could change the category boundaries. R1–R4 also overlap with architecture work pursued by the authors. To make these decisions inspectable, we publish group assignments, boundary cases, counterexamples, and the evidence trace; this documentation does not remove the risk of single-reviewer bias.

VII. Conclusion and Future Work

This review mapped 46 empirical studies of LLM-enabled embodied robotics. Direct generation and static structured approaches each account for 18 studies, formal planning or checking without autonomous feedback for four, and runtime adaptation or recovery for six. The additional full-text retrieval revealed recovery systems without formal symbolic verification and changed the original distribution across the taxonomy. It did not reveal a system combining semantic diagnosis of physical failure, runtime updates to formal constraints, and autonomous replanning in dynamic, unstructured environments.

The closest systems address different parts of that process: PaLM-E and open-world grasping revise behavior without a formal-constraint update, while RONAR couples failure interpretation to human-executed recovery. Future work can test whether semantic error reports and updated PDDL or LTL constraints support reliable replanning while fast safety control remains independent. The R1–R4 requirements offered here are candidate design choices that still require experimental validation.

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